ITC-OVAD: An Image–Text Collaborative Strategy for Open-Vocabulary Aerial Detection

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***Abstract*—Aerial object detection plays a crucial role in numerous applications. However, most existing methods focus on detecting predefined object categories, limiting their applicability in real-world open scenarios. In this article, we extend aerial object detection to open scenarios through image–text collaboration and propose ITC-OVAD, an end-to-end open-vocabulary detector for aerial imagery. Specifically, we first introduce an image-to-text alignment loss to replace the conventional category regression loss, thereby eliminating category constraints. Next, we propose a lightweight image–text collaboration strategy with bidirectional modulation between visual and textual modalities. Specifically, this strategy comprises three modules: text-guided feature enhancement (TG-FE) and visual-guided text refinement (VG-TR) in the encoder, and text-guided query enhancement (TG-QE) in the decoder, which together enrich visual features with class semantics and refine class embeddings with visual context, improving detection accuracy without incurring significant computational overhead. Extensive experiments demonstrate that ITC-OVAD consistently outperforms existing state-of-the-art (SOTA) methods across open-vocabulary, zero-shot, and traditional closed-set detection tasks. For instance, on the open-vocabulary aerial detection benchmarks DIOR, DOTA-v2.0, and LAE-80C, ITC-OVAD achieves 87.7 AP_{50}, 53.8 mean average precision (mAP), and 23.7 mAP, respectively, surpassing the previous SOTA (LAE-DINO) by 2.2, 7.0, and 3.5 points. Meanwhile, ITC-OVAD achieves real-time inference at 34 frames per second (FPS) on an RTX 4090 GPU, approximately three times faster than LAE-DINO (10 FPS). The code will be released at https://github.com/GT-Wei/ITC-OVAD**

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***Index Terms*—Aerial object detection, image–text collaboration, open-vocabulary detection, remote sensing, vision–language model.**

I. Introduction

AERIAL object detection, which involves localizing and categorizing objects of interest within aerial images, has become increasingly important due to its broad application requirements, including earth monitoring, disaster search, and rescue [1], [2], [3]. Driven by these practical demands, recent remote-sensing studies have explored efficient visual representation and target recognition across optical and SAR imagery [4], [5], [6], [7]. Nevertheless, most existing aerial detectors [8], [9], [10], [11], [12], [13], [14] can only identify predefined categories from their training sets. Such closed-set detectors require additional annotation and retraining when novel categories emerge, limiting their deployment in open scenarios. A major obstacle to discovering novel classes is their reliance on mapping image features to a fixed set of class indices [15].

Recently, CLIP [16] has the capability to map vision–language features into a comparable representation space. This offers an opportunity to break the category limitations inherent in traditional object detection. Inspired by this, several works [17], [18], [19], [20] on natural images have explored the relationship between images and text to endow models with open-vocabulary detection capabilities, achieving notable success. For instance, VILD [17] and VL-PLM [18] use a pretrained VLM to classify the object proposals generated by a region proposal network. GLIP [21] formulates object detection as a phrase grounding task, thereby unifying these two tasks. The introduction of textual information and the synergy between image and text have significantly expanded the coverage of traditional object detection tasks, taking them a step closer to real-world applications in open environments.

However, in the field of aerial object detection, such work remains limited. Aerial object detection poses unique challenges, including predominantly small objects, varying viewpoints, background interference, and limited detection data. Directly applying existing open-vocabulary detection methods designed for conventional perspectives to aerial perspectives may result in suboptimal performance. Very recently, some researchers have attempted to address this gap and proposed insightful methods [22], [23], [24], [25]. These works have demonstrated the potential of the open-vocabulary

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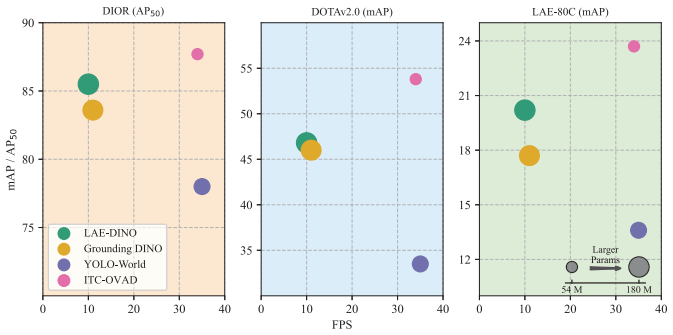


Fig. 1. Comparison of ITC-OVAD with other advanced open-vocabulary detectors on the DIOR, DOTA-v2.0, and LAE-80C datasets in terms of mAP/AP_{50}, FPS, and parameter count. Marker size indicates the number of parameters, with larger markers representing models with more parameters. All models are evaluated under the open-vocabulary aerial detection setting.

approach in the aerial object detection field, but there is still significant room for optimization. First, image–text alignment poses a significant challenge for open-vocabulary aerial detection. Existing methods often adopt alignment strategies developed for natural-image-based detectors, which may lead to overfitting due to the limited aerial detection data. Second, current open-vocabulary aerial detectors typically overlook visual–semantic interaction and, in particular, fail to address the background interference that arises during this process, which is especially prominent in aerial imagery. Finally, there remains a scarcity of efficient end-to-end open-vocabulary aerial detectors. Most existing methods [23], [24] utilize complex semi-supervised training strategies that heavily rely on pseudo-label generation for discovering novel classes. Although LAE-DINO [25] achieves an end-to-end framework based on Grounding DINO, it still incurs substantial inference cost, leaving room for a more efficient alternative.

In this article, we propose an end-to-end open-vocabulary detection model specifically designed for aerial imagery, termed ITC-OVAD. Specifically, we adopt the basic architecture of RT-DETR, due to its high inference efficiency and concise structure. First, we adapt the pretrained CLIP model and integrate it into the base framework to leverage its image–text alignment capability and powerful feature extraction, alleviating the limited sample problem. We then propose an image-to-text alignment loss to bridge the domain gap between natural images and aerial images, extending the model's applicability from closed-world to open-world scenarios. Next, considering the unique challenges of aerial imagery, such as weak features from small targets and strong background interference, we propose a lightweight image–text collaboration strategy with bidirectional modulation between visual and textual modalities. Specifically, this strategy comprises three modules: text-guided feature enhancement (TG-FE) and visual-guided text refinement (VG-TR) in the encoder, and text-guided query enhancement (TG-QE) in the decoder. These modules jointly enhance visual and textual representations while suppressing background interference, further improving detection accuracy without incurring substantial computational overhead. Finally, we conduct comprehensive experiments across three benchmark tasks: open-vocabulary detection, zero-shot detection (ZSD), and traditional closed-set detection. As illustrated in Fig. 1, ITC-OVAD consistently surpasses mainstream open-vocabulary detectors (Grounding DINO and YOLO-World) as well as the recent aerial-specific detector LAE-DINO in terms of both mean average precision (mAP) and inference speed, demonstrating the effectiveness of the proposed approach.

In summary, there are three major contributions.

1) We propose an image-to-text alignment loss as a bridge to address the modality gap between natural images and aerial images. By integrating it with CLIP, we extend the RT-DETR framework from closed-world to open-world scenarios.
2) We propose a lightweight image–text collaboration strategy with bidirectional modulation between visual and textual modalities. Specifically, it comprises three modules: TG-FE and VG-TR in the encoder, and TG-QE in the decoder. These modules jointly enhance visual and textual representations while suppressing background interference, further improving detection accuracy without significant computational overhead.
3) Our method achieves state-of-the-art (SOTA) performance across multiple benchmark datasets for open-vocabulary and zero-shot aerial detection, as well as traditional closed-set tasks, while maintaining real-time inference at 34 FPS.

II. Related Work

A. Open-Vocabulary Detection in Natural Images

Traditional object detectors [26], [27], [28], [29] predominantly rely on fully supervised training and map image features to predefined class indices [15]. Consequently, they can recognize only the categories seen during training, a setting commonly referred to as fixed-vocabulary or closed-set detection. Open-vocabulary object detection (OVD), in contrast, exploits image–text relationships to recognize categories beyond the training vocabulary. Leveraging the image–text alignment learned by CLIP [16], existing OVD methods introduce vision–language knowledge into object detectors through two main routes. The first route transfers such knowledge to region-level detectors through distillation, pseudo-labeling, or region–text pretraining [17], [18], [30], [31], [32], [33]. For example, ViLD [17] distills knowledge from a pretrained vision–language model, whereas VL-PLM [18] generates region–text pseudo-labels from unlabeled images to train the detector. The second route unifies object detection with phrase grounding and jointly trains vision–language detectors using large-scale image–text data [19], [20], [21], [34], [35]. GLIP [21] formulates detection and grounding within a unified framework, while Grounding DINO [19] extends the DINO detector with language-guided query selection and cross-modal feature fusion for effective image–text interaction during detection. The former route depends on region proposals and additional knowledge-transfer procedures, whereas the latter generally requires large-scale grounding data.

The most closely related natural-image methods are YOLO-World [20] and Grounding DINO [19]. YOLO-World performs image–text interaction in RepVL-PAN, while Grounding

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DINO introduces cross-modal fusion into feature enhancement, query selection, and decoding. Both methods learn cross-modal alignment during detector pretraining using large-scale natural-image detection and grounding data. In contrast, ITC-OVAD targets aerial imagery and uses the aligned CLIP visual and text encoders as initialization. It preserves the pretrained text space through image-to-text alignment, while the soft foreground gates in TG-FE and TG-QE regulate the textual information received at different spatial locations and object queries to reduce background interference.

B. Closed-Set Aerial Object Detection

Closed-set aerial object detection localizes objects and assigns them to categories from a fixed set defined before training. Existing studies mainly address the inherent characteristics of remote-sensing imagery, including small and densely distributed objects, large variations in scale and orientation, and extensive background interference. ClusDet, DMNet, and UFPMP-Det adopt coarse-to-fine or region-refinement strategies to focus detection on object-dense regions [9], [10], [11].

Beyond these region-focusing strategies, recent work has improved multiscale representation and computational efficiency [8], [12], [13], as well as oriented localization and fine-grained discrimination [36], [37], [38]. Related research also covers efficient visual analysis in optical imagery [4], [5] and target detection and recognition in SAR imagery [6], [7]. Although these methods address important characteristics of aerial imagery, they cannot directly recognize categories outside the training vocabulary. This limitation motivates the shift from closed-set to open-vocabulary aerial detection.

C. Open-Vocabulary Aerial Detection

Recent studies introduce vision–language knowledge into aerial detectors through different technical routes [22], [23], [24], [25]. DescReg [22] uses visual descriptions and a similarity-aware triplet loss to regularize the classification space, but does not introduce image–text interaction throughout the detector. CastDet [23], [24] adopts a student–teacher framework to generate proposals and pseudo-labels for novel categories, but requires multiple teachers and iterative pseudo-label updates. From a data and pretraining perspective, LAE-DINO [25] constructs the LAE-1M dataset and adapts Grounding DINO using dynamic vocabularies and visual-guided text prompts, yet inherits the heavy architecture.

In parallel, remote-sensing vision–language models, such as GeoChat, LHRS-Bot, and EarthGPT [39], [40], [41], explore general-purpose multimodal understanding of remote-sensing imagery. They support a broad range of tasks, including visual question answering, image captioning, visual grounding, and multisensor image interpretation. Their primary objective differs from that of ITC-OVAD: these models pursue unified multimodal understanding and reasoning, typically with multibillion-parameter language backbones. ITC-OVAD instead focuses on efficient end-to-end OVD, combining stable image-to-text alignment with lightweight image–text collaboration in the encoder and the decoder.

III. Method

A. Model Architecture

The overall architecture of ITC-OVAD is illustrated in Fig. 2. ITC-OVAD consists of two main components.

1) Image-to-text alignment establishes a shared vision–language representation space as the foundation for open-vocabulary detection. It freezes the CLIP text encoder and learns a visual-side projection into the CLIP text feature space, thereby removing the predefined-category constraint while preserving CLIP's pretrained semantic generalization to novel classes.
2) Image–text collaboration employs bidirectional interaction: text-guided modules sharpen visual features around class-relevant cues, while visual-guided refinement enriches class semantics with contextual evidence. Concretely, we introduce three lightweight modules: TG-FE and VG-TR in the encoder, and TG-QE in the decoder. These modules jointly address the challenges of open-vocabulary detection in aerial imagery, such as small targets with weak visual features and strong background interference.

1) Image-to-Text Alignment Loss: Image–text alignment commonly applies learnable projections to both visual and text features. Let q_i denote the ith object query, t_c^0 denote the pretrained CLIP embedding of class c, and W_V and W_T are the visual- and text-side projections, respectively. Dual-side alignment and the proposed Image-to-text alignment can be written as

$$\begin{aligned} s_{ic}^{\text{dual}} &= \text{sim}\left(W_V q_i, W_T t_c^0\right), \quad \Theta_{\text{dual}} = \{W_V, W_T\} \\ s_{ic}^{\text{I2T}} &= \text{sim}\left(W_V q_i, t_c^0\right), \quad \Theta_{\text{I2T}} = \{W_V\} \end{aligned} \tag{1}$$

where $\text{sim}(\cdot,\cdot)$ denotes cosine similarity, and Θ denotes the projection parameters learned during alignment. In aerial ZSD, instance-level detection supervision is available only for base classes, while textual supervision usually consists of a small number of category names. The text-side projection W_T is therefore learned only through base classes, without direct constraints on the projected representations of novel classes, which may change the relative relations inherited from CLIP. We instead freeze the CLIP text encoder, apply no additional learnable projection to its output embeddings, and learn only a visual-side projection into the CLIP text feature space. This allows the visual features to adapt to instance-level detection supervision while preserving the pretrained class semantics.

a) Contrastive head: The contrastive head implements this unidirectional projection. As illustrated in Fig. 2, it takes two inputs: the object query embeddings $\mathbf{v}$ from the text-guided decoder, and the refined class embeddings $\mathbf{t}$ from the VG-TR module (Section III-B1.b). A single linear layer maps the visual features into the text embedding space, while no additional projection is applied to the refined class embeddings within the contrastive head. The visual-text similarity $S(\mathbf{v},\mathbf{t})$ is then computed as

$$S(\mathbf{v},\mathbf{t}) = \tau \frac{\text{proj}(\mathbf{v}) \cdot \mathbf{t}^T}{\|\text{proj}(\mathbf{v})\|_2 \cdot \|\mathbf{t}\|_2} + \beta \tag{2}$$

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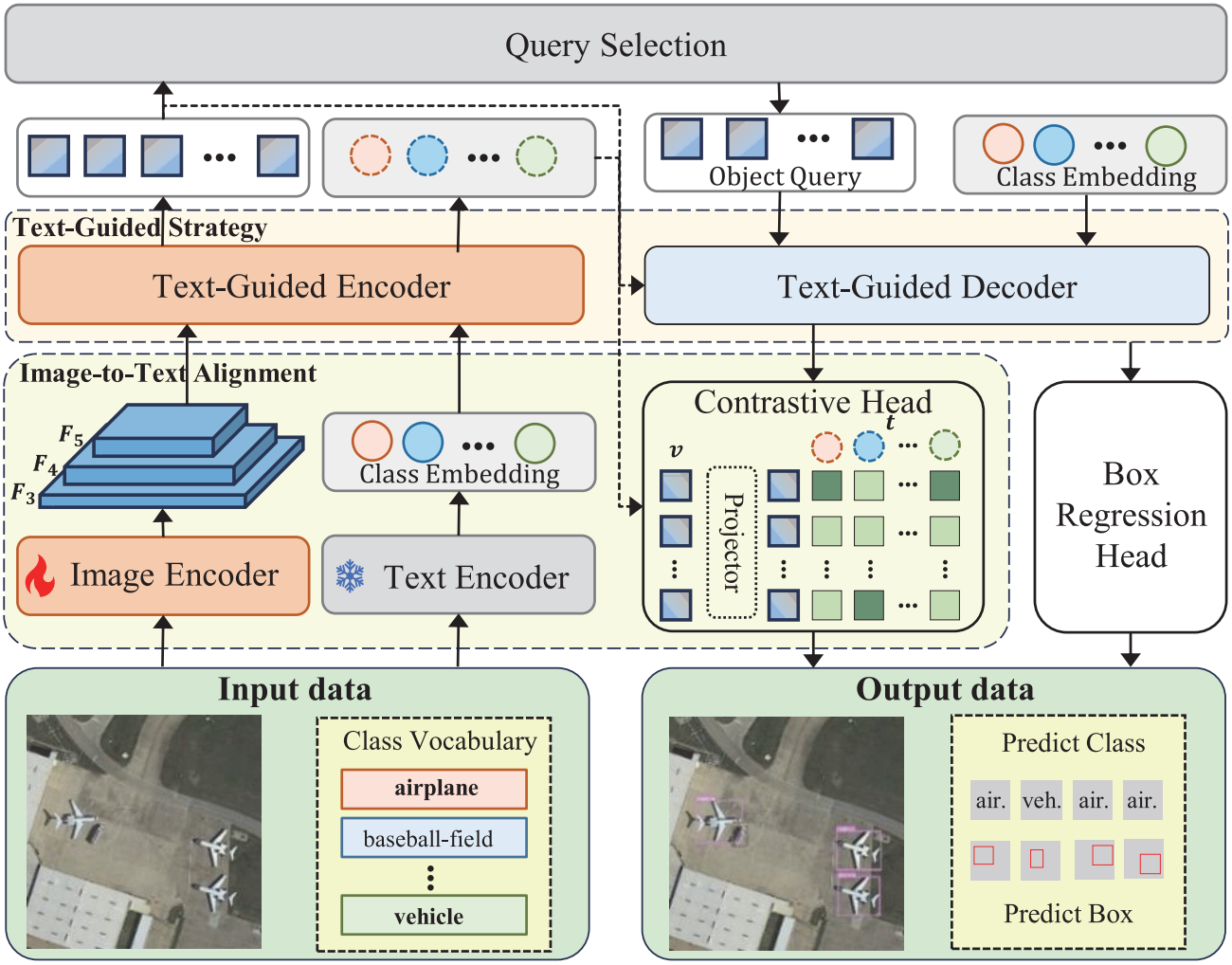


Fig. 2. Overall architecture of ITC-OVAD. The proposed ITC-OVAD can be summarized into two main components: image-to-text alignment and image–text collaboration strategy: 1) image-to-text alignment connects visual objects with class embeddings rather than relying on class indices, which enables the model to detect classes beyond a predefined set. It involves a text encoder, a contrastive head, and an image-to-text alignment loss, and 2) the image–text collaboration strategy explores cross-modal interaction with bidirectional modulation between visual and textual modalities, enhancing class-relevant feature extraction and suppressing background interference, forming an image–text collaboration encoder–decoder structure.

where proj(·) is a bias-enabled fully connected layer that maps each 256-D decoder query to the 1024-D CLIP text feature space. This projection is used only in the classification branch to compute visual–text similarities, while the box regression branch uses the decoder query before projection. The similarity scale τ and bias β are learnable and initialized to $1/0.07$ and 0, respectively.

b) Detection loss: The detection loss consists of two terms: an image-to-text alignment loss L_{con} and a location loss. Given an image and a class vocabulary of N_c classes, ITC-OVAD outputs N_q object predictions $\{(b_i, s_i)\}_{i=1}^{N_q}$ per decoder layer, where N_q is the number of queries, with corresponding ground-truth labels $\{(b_m, u_m)\}_{m=1}^{M}$, and M is the number of ground-truth objects.

The alignment loss L_{con} replaces the traditional classification loss with three components that align it with detection's query-region supervision: 1) a pointwise contrastive formulation between each query and the class embeddings; 2) sigmoid normalization [42] to decouple per-class gradients from class and batch dependencies; and 3) varifocal-style intersection over union (IoU)-weighted targets [43], where each prediction's contribution is scaled by its predicted-to-ground-truth IoU. This combination matches detection's query-region pairing rather than image-caption matching and couples classification confidence with localization quality through the IoU-weighted target. We formally define L_{con} as

$$\mathrm{L_{con}}(\theta_c, u_c) = \begin{cases} -u_c \left[u_c \log(\theta_c) \right. \\ \qquad \left. + (1-u_c) \log(1-\theta_c) \right], & \text{if } u_c > 0 \\ -\alpha\, \theta_c^{\gamma} \log(1-\theta_c), & \text{if } u_c = 0 \end{cases} \tag{3}$$

where $\theta_c = \sigma(s_c)$ is the sigmoid score for class c. For a matched query, the target u_c is set to the IoU between the predicted and ground-truth boxes if c is the ground-truth class, and to 0 otherwise. For an unmatched query, $u_c = 0$ for all classes. The balance weight α and focusing exponent γ are set to 0.75 and 2.0, respectively, following the default settings of the base framework.

Following previous works [29], [44], we adopt the GIoU loss L_{GIoU} and L1 loss L_{L1} for location regression. The total

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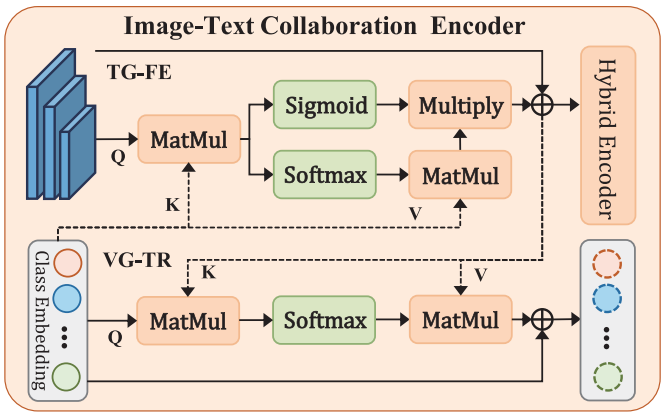


Fig. 3. Image–text collaboration encoder in ITC-OVAD. The TG-FE module enhances feature extraction by incorporating relevant class semantics into the visual features. The VG-TR module further refines the textual representations by fusing the enhanced visual features from TG-FE into the corresponding class embeddings.

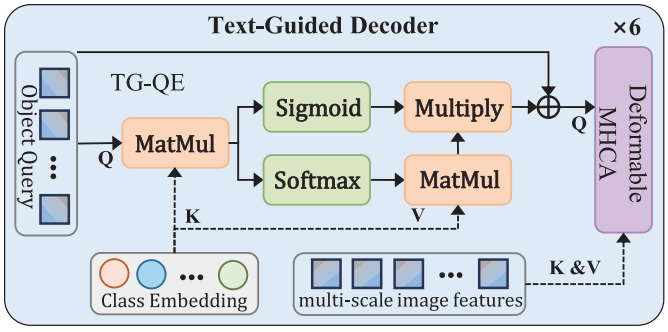


Fig. 4. Text-guided decoder in ITC-OVAD. By incorporating class information into object queries, TG-QE enables more focused retrieval of class-relevant features.

training loss is

$$L_{det} = \lambda L_{con} + \mu L_{GIoU} + \nu L_{L1} \tag{4}$$

where λ, μ, and ν are the loss weights for the alignment, GIoU, and L1 losses, and are set to 1, 2, and 5, respectively.

B. Image–Text Collaboration Strategy

Through the image-to-text alignment introduced above, we have already extended detection beyond predefined categories. However, the limited training data and unique challenges inherent in aerial imagery, such as small targets producing weak visual cues and the broad coverage of aerial images introducing substantial background interference, still pose difficulties for effective image–text alignment.

To mitigate these difficulties, we propose an image–text collaboration strategy with bidirectional modulation between the two modalities. Text-guided modulation enriches visual features with class semantics, while visual-guided modulation refines class embeddings with visual context. We implement this collaboration through three lightweight modules: 1) in the encoder, TG-FE and VG-TR form a feature-level bidirectional collaboration and 2) in the decoder, TG-QE extends text-guided modulation to the query level. The two directions are complementary, jointly strengthening image–text alignment.

1) Image–Text Collaboration Encoder:

a) Text-guided feature enhancement: As noted above, aerial imagery presents two visual-side challenges to image–text alignment: 1) due to the long imaging distance, aerial images often contain small, low-resolution targets whose visual features are not prominent and 2) due to the broad scene coverage, aerial images often contain extensive backgrounds that strongly interfere with the detection process. To address both, we propose TG-FE to enhance the encoder's visual features with class semantics.

As illustrated in Fig. 3, the core idea is based on cross-attention. Specifically, given multiscale image features $F_i \in \mathbb{R}^{H\times W\times C}$ from the image encoder and class embeddings $T \in \mathbb{R}^{N_c\times D}$, where N_c is the number of classes, we employ cross-attention to fuse these modalities. To suppress the excessive injection of class embeddings into background features, we compute the maximum similarity between each image feature and the class embeddings, then apply a sigmoid function to interpret this as the probability that the image feature belongs to the foreground. We then multiply the text feature representation $T_j^{(v)}$ by this probability. The image features are updated as follows:

$$\text{Attn}_{\text{FE}} = \frac{F_i^{(q)} T^{(k)\mathsf{T}}}{\sqrt{d}} \tag{5}$$

$$F_i' = F_i + \phi\left(\text{Attn}_{\text{FE}}\right) \cdot T^{(v)} \cdot \sigma\left(\max_{j=1,\ldots,N_c}\left(\text{Attn}_{\text{FE}}\right)\right) \tag{6}$$

where $F_i^{(q)}$ denotes the query projection of the ith layer image feature in the multiscale features, $T^{(k)}$ and $T^{(v)}$ represent the key and value projections of the class-embedding matrix, respectively, j indexes the classes, $\phi(\cdot)$ denotes the Softmax function, $\sigma(\cdot)$ denotes the Sigmoid function, and d denotes the channel dimension of the projected query and key features. Additionally, to reduce the parameter count and ensure a lightweight design, features of different scales share the same set of TG-FE parameters.

The Softmax branch integrates text information into the image features, which can improve issues related to inconspicuous target visual features. Meanwhile, the sigmoid branch, guided by text information, functions like a gating mechanism, suppressing irrelevant background interference during detection.

b) Visual-guided text refinement: While TG-FE enhances visual features with textual guidance, the class embeddings remain static. Due to varying viewpoints and shooting altitudes, objects of the same category often exhibit diverse visual appearances in aerial images, making it difficult to align them with a single fixed class embedding. A natural solution is to enrich class embeddings with visual features. However, directly fusing raw visual features risks introducing background interference into class representations.

To address this, we introduce a VG-TR module following TG-FE. Specifically, VG-TR employs a cross-attention mechanism to fuse the enhanced visual features produced by TG-FE with the textual embeddings. Formally, given the

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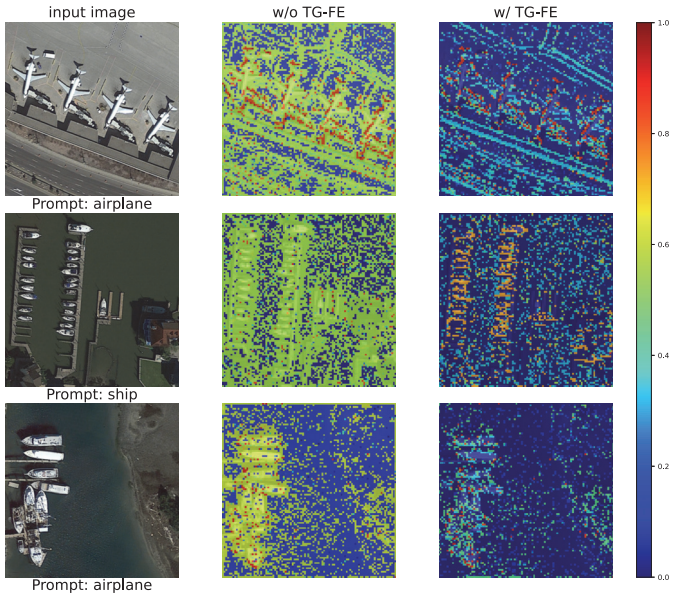


Fig. 5. Impact of the TG-FE module on features. The first column displays the input images, while the subsequent two columns illustrate the correlation between the image features and the detected targets.

enhanced visual features $F_i' \in \mathbb{R}^{H\times W\times C}$ generated by the TG-FE module, and the class-embedding matrix $T \in \mathbb{R}^{N_c\times D}$, the VG-TR module updates the class embeddings as follows:

$$\text{Attn}_{\text{TR}} = \frac{T^{(q)} F_i'^{(k)\text{T}}}{\sqrt{d}} \tag{7}$$

$$T' = T + \phi\left(\text{Attn}_{\text{TR}}\right) \cdot F_i'^{(v)} \tag{8}$$

where $T^{(q)}$ denotes the query projection of the class-embedding matrix, $F_i'^{(k)}$ and $F_i'^{(v)}$ represent the key and value projections of the refined visual features from TG-FE, respectively, and $\phi(\cdot)$ denotes the Softmax function.

TG-FE first uses the class embeddings to enhance the visual features, with foreground gating reducing text injection into background regions. VG-TR then takes the resulting features F_i', rather than the original features F_i, as the key and value to update the class embeddings. Thus, the class embeddings guide visual feature enhancement in TG-FE, while the enhanced visual features provide image-specific cues for text refinement in VG-TR.

As a result, the generated text embeddings incorporate class-relevant visual cues and contextual scene information, producing class representations that better adapt to the diverse visual appearances of objects in aerial scenes. Ultimately, these refined and context-aware textual embeddings are utilized to compute visual-text similarity within the contrastive head.

2) Text-Guided Query Enhancement: In the decoder, object queries retrieve class-relevant features from the enriched encoder outputs through cross-attention. As class embeddings guide visual features through TG-FE at the encoder, they can also guide object queries toward more focused retrieval of class-relevant features at the decoder. As shown in Fig. 4, TG-QE injects class embeddings into object queries as prior knowledge, so that their subsequent interaction with multiscale image features in the decoder concentrates on class-relevant regions.

However, the number of object queries is typically much larger than the number of actual objects, leaving most queries as negative samples. Indiscriminately injecting class embeddings into these negative queries may make them resemble positive samples and interfere with label assignment. TG-QE thus adopts the same sigmoid gating as in TG-FE, adaptively suppressing class injection for likely negative queries. Specifically, the object queries are updated as follows:

$$\text{Attn}_{\text{QE}} = \frac{Q_i^{(q)} T_j^{(k)T}}{\sqrt{d}} \tag{9}$$

$$Q_i' = Q_i + \phi\left(\text{Attn}_{\text{QE}}\right) \cdot T_j^{(v)} \cdot \sigma\left(\max_{j=1,\ldots,N_c}\left(\text{Attn}_{\text{QE}}\right)\right) \tag{10}$$

where $Q_i^{(q)}$ denotes the query projection of the object queries at the ith decoder layer, $T_j^{(k)}$ and $T_j^{(v)}$ represent the key and value projections of the text embeddings, $\phi(\cdot)$ denotes the Softmax function, and $\sigma(\cdot)$ denotes the Sigmoid function. The same TG-QE parameters are shared across all six decoder layers.

IV. Experiments

In this section, we aim to demonstrate that the proposed ITC-OVAD has the ability to break category limitations and explicitly demonstrate the effectiveness of our proposed components. To comprehensively evaluate our method, we conduct experiments on three benchmark tasks: open-vocabulary aerial object detection, zero-shot aerial detection, and traditional closed-set object detection.

A. Datasets and Experiment Setup

1) Datasets: For the evaluation of open-vocabulary aerial object detection, we follow the benchmark protocol established in [25], where the model is first pretrained on LAE-1M and then evaluated on three benchmark datasets: DIOR [1], DOTA-v2.0 [45], and LAE-80C [25].

For the evaluation of zero-shot aerial object detection, we use three widely adopted benchmark datasets: xView [46], DIOR [1], and DOTA [45]. To ensure fairness and validity in our comparative experiments, we propose two distinct protocols for splitting the datasets into base and novel classes. In the first protocol, we follow the setup proposed in DescReg [22]. However, we observe that some novel classes in DIOR overlap with the base classes in DOTA, and that DOTA-v1.0 lacks sufficient annotations for small instances. To avoid class leakage and mitigate annotation inconsistencies, in the second protocol we adopt DOTAv1.5 and adjust the category splits in DOTA accordingly. Specifically, Protocol 2 retains the original DIOR split and revises only the DOTA split: basketball court and ground-track-field are moved from base to novel, while tennis court and soccer-ball-field are moved from novel to base. Full details of the base/novel category splits used for zero-shot evaluation are provided in Supplementary Material.

For conventional closed-set aerial object detection, ITC-OVAD is further evaluated on two standard UAV benchmarks: VisDrone [47] and UAVDT [48].

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TABLE I

COMPARISON WITH SOTA OPEN-VOCABULARY DETECTORS ON DIOR, DOTAV2.0, AND LAE-80C BENCHMARKS. ALL MODELS ARE PRETRAINED ON LAE-1M. ALL FPS VALUES ARE UNIFORMLY MEASURED ON THE SAME RTX 4090 GPU. THE BEST RESULT WITHIN EACH SPLIT IS SHOWN IN BOLD, AND THE SECOND-BEST IS UNDERLINED

Method	Source	Pre-training Data	Resolution	Params	FPS	DIOR	DOTAv2.0	LAE-80C
						AP_{50}	mAP	mAP
YOLO-World [20]	CVPR 2024	LAE-1M	800×800	110M	35	78.0	33.5	13.6
Grounding DINO [19]	ECCV 2024	LAE-1M	1333×800	172M	11	83.6	46.0	17.7
LAE-DINO [25]	AAAI 2025	LAE-1M	1333×800	180M	10	<u>85.5</u>	<u>46.8</u>	<u>20.2</u>
ITC-OVAD (Ours)	—	LAE-1M	800×800	54M	34	**87.7** (+2.2)	**53.8** (+7.0)	**23.7** (+3.5)

2) Evaluation Metrics: Following [22] and [25], we adopt standard object detection metrics for evaluation, including mAP, AP_{50}, recall, harmonic mean (HM), and FPS. The mAP measures the average precision across multiple IoU thresholds ranging from 0.5 to 0.95 (in steps of 0.05). AP_{50} and recall are computed using an IoU threshold of 0.5. The HM metric is used to reflect the overall performance across base and novel classes, and is computed as the HM of their respective scores. Additionally, for ZSD tasks, we report results under two evaluation settings: ZSD, which evaluates performance on novel classes only, and generalized ZSD (GZSD), which assesses performance on both base and novel classes simultaneously.

3) Implementation Details: ITC-OVAD is implemented based on RT-DETR [29] with a ResNet50 backbone [49]. By incorporating an additional text encoder and contrastive loss, we form an end-to-end detection framework. All experiments were conducted on four 4090 GPUs, each with a batch size of 2. We set the number of object queries to 500 to accommodate the higher number of objects typically found in aerial images compared to natural scenes. Unless otherwise specified, the FPS are reported for a single 4090 GPU without the use of any additional acceleration techniques. Due to the space limitation, further details on the implementation are provided in the Supplementary Material.

B. Performance on Open-Vocabulary Aerial Detection

As shown in Table I, we evaluate the proposed ITC-OVAD on the open-vocabulary aerial detection benchmark established by [25], achieving SOTA performance across three benchmark datasets. Specifically, we compare our method with two mainstream open-vocabulary detectors designed for natural images (Grounding DINO and YOLO-World), as well as the recent aerial-specific detector LAE-DINO. The experimental results demonstrate that, despite using a lower-resolution input of 800 × 800, our ITC-OVAD still significantly outperforms previous SOTA methods, surpassing them by +2.2, +7.0, and +3.5 in terms of AP_{50} on the DIOR dataset, and mAP on DOTAv2.0 and LAE-80C datasets, respectively.

1) Model Size and Latency: Table I also compares model complexity and inference speed. In particular, compared to other methods, ITC-OVAD uses fewer parameters while achieving the highest AP_{50}/mAP performance on three benchmark datasets. ITC-OVAD also achieves an inference speed of 34 FPS, notably faster than Grounding DINO (11 FPS) and LAE-DINO (10 FPS), and comparable to YOLO-World (35 FPS). These results highlight that ITC-OVAD successfully inherits the efficiency of its base architecture, making it particularly well-suited for open scenarios and aerial detection.

C. Performance on Zero-Shot Aerial Detection

As mentioned in Section IV-A, we adopt two protocols for splitting the base and novel classes. The second protocol is designed to avoid class leakage, and its results are marked with an asterisk (*) in the subsequent tables. For a comprehensive evaluation, we compare our approach not only with SOTA aerial zero-shot detectors but also with open-vocabulary detection methods YOLO-World [20], Grounding DINO [19] and LAE-DINO [25].

1) Evaluation on xView: In Table II, we compare the performance of ITC-OVAD with SOTA methods on the xView dataset. Under the ZSD setting, ITC-OVAD achieves the highest AP_{50} of 9.8% and recall of 60.1% in the first protocol, outperforming Grounding DINO (+1.1% AP_{50}, +1.5% recall), LAE-DINO (+2.6% AP_{50}, +1.8% recall), and YOLO-World (+1.9% AP_{50}, +23.0% recall). In the second protocol, ITC-OVAD achieves an AP_{50} of 12.3%, surpassing all other compared methods.

In the GZSD setting, ITC-OVAD consistently outperforms all compared methods in recall across both base and novel classes. It achieves a novel-class AP_{50} of 7.0% and recall of 36.7% in the second protocol, achieving the highest HM in both AP and recall.

2) Evaluation on DIOR and DOTA: We further evaluate ITC-OVAD on the DIOR and DOTA datasets under both ZSD and GZSD settings, as shown in Table III. On DIOR, ITC-OVAD* achieves the highest ZSD AP_{50} of 14.1%, surpassing LAE-DINO* and Grounding DINO* by 3.7% and 4.3%, respectively. In terms of recall, ITC-OVAD* reaches 81.2%, outperforming all compared methods and demonstrating strong generalization to novel classes. In the GZSD setting, ITC-OVAD also achieves the highest HM of AP_{50} at 13.9%, representing a +1.9% improvement over LAE-DINO*. On the DOTA dataset, ITC-OVAD* achieves the best ZSD AP_{50} of 9.6% and competitive recall. While ITC-OVAD exhibits slightly lower performance than LAE-DINO in a few GZSD metrics, it demonstrates clear advantages in model efficiency: as shown in Table I, ITC-OVAD significantly reduces parameter count and inference latency compared to both Grounding DINO and LAE-DINO. These results underscore that ITC-OVAD offers more robust open-vocabulary detection capabilities in aerial detection.

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TABLE II

COMPARISON WITH THE SOTA UNDER GZSD AND ZSD ON THE XVIEW DATASET, WHERE * DENOTES THAT THE DATASET SPLIT REDEFINES THE BASE AND NOVEL CLASSES TO AVOID LEAKAGE OF NOVEL CLASSES. THE BEST RESULT WITHIN EACH SPLIT IS SHOWN IN BOLD, AND THE SECOND-BEST IS UNDERLINED

Method	Source	Backbone	Resolution	GZSD						ZSD	
				AP_{50}			Recall			AP_{50}	Recall
				Base	Novel	HM	Base	Novel	HM		
RRFS [50]	CVPR 2022	ResNet–101	800×800	10.2	1.6	2.7	19.1	5.8	8.9	2.2	14.3
ContrastZSD [51]	TPAMI 2022	ResNet–101	800×800	16.8	2.9	5.0	27.6	13.9	18.5	4.1	27.1
DescReg [22]	AAAI 2024	ResNet–101	800×800	17.1	**5.8**	**8.7**	28.0	12.8	17.6	8.3	43.0
YOLO-World [20]	CVPR 2024	YOLOv8-L	800×800	18.5	3.3	5.6	41.4	15.2	22.2	7.9	37.1
Grounding DINO [19]	ECCV 2024	Swin-T	1333×800	20.3	4.8	7.7	47.0	22.1	30.0	8.7	58.6
LAE-DINO [25]	AAAI 2025	Swin-T	1333×800	20.5	4.7	7.6	46.2	31.2	37.3	7.2	58.3
ITC-OVAD (Ours)	–	ResNet–50	800×800	**21.1**	5.3	8.5	**58.9**	**34.9**	**43.8**	**9.8**	**60.1**
YOLO-World*	CVPR 2024	YOLOv8-L	800×800	21.7	3.2	5.6	46.6	16.5	24.4	8.8	41.7
Grounding DINO*	ECCV 2024	Swin-T	1333×800	20.9	6.2	9.5	47.8	31.1	37.7	11.1	**63.5**
LAE-DINO*	AAAI 2025	Swin-T	1333×800	21.4	4.6	7.6	50.1	26.0	34.0	8.9	58.8
ITC-OVAD (Ours)*	–	ResNet–50	800×800	**22.4**	**7.0**	**10.6**	**60.1**	**36.7**	**45.5**	**12.3**	62.5

TABLE III

COMPARISON WITH THE SOTA UNDER GZSD AND ZSD ON THE DIOR AND DOTA DATASETS. TWO EVALUATION PROTOCOLS ARE REPORTED: THE UPPER BLOCK ADOPTS THE STANDARD BASE/NOVEL SPLIT, WHILE THE LOWER BLOCK (MARKED WITH *) FOLLOWS A REDEFINED SPLIT TO AVOID NOVEL-CLASS LEAKAGE. THE BEST VALUE IS SHOWN IN BOLD AND THE SECOND-BEST IS UNDERLINED

Method	DIOR								DOTA							
	GZSD						ZSD		GZSD						ZSD	
	AP_{50}			Recall			AP_{50}	Recall	AP_{50}			Recall			AP_{50}	Recall
	Base	Novel	HM	Base	Novel	HM			Base	Novel	HM	Base	Novel	HM		
RRFS	41.9	2.8	5.2	60.0	19.9	29.9	9.7	19.8	47.1	2.2	4.2	71.4	14.2	23.7	2.9	14.4
ContrastZSD	51.4	3.9	7.2	69.2	25.9	37.7	8.7	22.3	41.6	2.8	5.2	69.1	12.2	20.7	6.0	25.4
DescReg	68.7	7.9	14.2	82.0	34.3	48.4	15.2	34.6	68.7	4.7	8.8	83.8	29.9	44.0	8.5	34.4
YOLO-World-L	80.2	17.3	28.5	91.1	38.2	53.8	24.8	37.8	74.4	22.7	34.8	87.8	42.2	57.0	24.3	35.7
Grounding DINO	**80.7**	29.4	43.1	94.1	**63.8**	**76.0**	37.7	**84.4**	**77.5**	23.5	36.1	92.8	65.4	76.7	28.1	76.4
LAE-DINO	79.4	**32.2**	**45.9**	93.5	63.5	75.7	**38.2**	74.9	76.8	22.9	35.3	91.5	56.4	69.7	24.2	73.1
ITC-OVAD	80.0	23.2	36.0	**94.6**	62.5	75.2	37.5	74.2	76.3	**24.0**	**36.5**	**93.8**	**74.1**	**82.8**	**32.4**	**80.3**
YOLO-World-L*	78.5	3.1	5.9	91.5	18.6	30.9	6.7	20.6	71.5	1.7	3.3	80.9	27.1	40.6	4.2	19.5
Grounding DINO*	78.4	6.0	11.2	93.9	65.3	77.0	9.8	81.0	71.2	2.8	5.4	**86.4**	47.8	61.6	7.5	75.0
LAE-DINO*	**79.0**	6.6	12.0	93.8	**66.5**	**77.8**	10.4	80.7	71.9	**4.8**	**9.0**	**86.4**	51.7	64.6	8.0	**76.7**
ITC-OVAD*	78.5	**7.6**	**13.9**	**94.2**	64.9	76.8	**14.1**	**81.2**	**72.2**	2.9	5.6	85.9	**66.6**	**74.9**	**9.6**	76.6

TABLE IV

COMPARISON WITH THE SOTA ON THE VISDRONE VALIDATION SET. "800–1200" INDICATES MULTISCALE INPUTS. THE BEST VALUE IS SHOWN IN BOLD AND THE SECOND-BEST IS UNDERLINED

Method	Backbone	Resolution	AP_{50}
DMNet [10]	ResNet-50	1000×600	47.6
QueryDet [52]	ResNet-50	2400×2400	48.1
ClusDet [9]	ResNet-50	1000×600	50.6
CZ Det [14]	ResNet-50	800–1200	58.3
RT-DETR [29]	ResNet-50	800×800	58.9
UFPMP-Det [11]	ResNet-50	1333×800	62.4
GFL V1 (CEASC) [8]	ResNet-18	1333×800	50.7
YOLO-World [20]	YOLOv8-L	800×800	45.0
YOLO-World [20]	YOLOv8-L	1280×1280	55.3
LAE-DINO [25]	Swin-T	1333×800	56.4
Grounding DINO [19]	Swin-T	1333×800	58.1
ITC-OVAD	ResNet-50	800×800	60.1
ITC-OVAD	ResNet-50	1280×1280	**64.6**

D. Performance on Traditional Aerial Detection

1) Performance on the VisDrone Dataset: Although ITC-OVAD is designed for open-vocabulary scenarios, it can also be applied to traditional aerial object detection. We conduct experiments on VisDrone to assess its closed-set performance, with results reported in Table IV. For a fair comparison with previous work, we test using different input resolutions. At an input size of 800 × 800, ITC-OVAD achieves an AP_{50} of 60.1%, an improvement of 1.2% compared to RT-DETR. Increasing the resolution to 1280 × 1280 boosts AP_{50} by 2.2% beyond the previously best-reported result, achieving SOTA performance.

2) Performance on the UAVDT Dataset: In addition to VisDrone, we further evaluate ITC-OVAD on the UAVDT [48] dataset. As shown in Table V, ITC-OVAD achieves the highest AP_{50}, surpassing the best-compared method by 1.0 point. In summary, ITC-OVAD not only adapts to open-vocabulary aerial object detection scenarios but also retains strong performance in closed settings.

E. Ablation Study

We conduct ablation experiments on the DIOR dataset to assess the contribution of each proposed component, as shown in Table VI. The baseline incorporates a text encoder and a vanilla contrastive loss for image–text alignment, which enables ZSD of novel classes.

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TABLE V

COMPARISON WITH THE SOTA ON THE UAVDT DATASET. THE BEST VALUE IS SHOWN IN BOLD AND THE SECOND-BEST IS UNDERLINED

Method	Backbone	Resolution	AP_{50}
DMNet [10]	ResNet-50	1000×600	24.6
ClusDet [9]	ResNet-50	1000×600	26.5
GLSAN [53]	ResNet-50	1000×600	28.1
ARMNet [54]	ResNet-50	1500×800	30.4
UFPMP-Det [11]	ResNet-50	1000×600	38.7
RT-DETR [29]	ResNet-50	800×800	<u>39.5</u>
GFL V1 (CEASC) [8]	ResNet-18	1024×540	30.9
YOLO-World [20]	YOLOv8-L	800×800	35.8
Grounding DINO [19]	Swin-T	1333×800	36.0
LAE-DINO [25]	Swin-T	1333×800	36.5
ITC-OVAD	ResNet-50	800×800	**40.5**

TABLE VI

ABLATION STUDY OF EACH COMPONENT ON THE DIOR DATASET. C-WL DENOTES LOADING THE CLIP ENCODER WEIGHTS; I2T LOSS DENOTES VISUAL-SIDE ALIGNMENT TO THE CLIP TEXT FEATURE SPACE WITHOUT AN ADDITIONAL TEXT-SIDE PROJECTION

I2T Align		I-T Collaboration Strategy			GZSD	ZSD
C-WL	I2T Loss	TG-FE	TG-QE	VG-TR	HM	AP_{50}
					2.5	3.4
✓					5.6	7.0
✓	✓				8.8	9.8
✓	✓	✓			10.4	12.3
✓	✓		✓		11.5	11.3
✓	✓	✓	✓		13.1	**14.3**
✓	✓	✓	✓	✓	**13.9**	14.1

1) Ablation on Image-to-Text Alignment: To facilitate image–text contrastive learning, we initialize both the visual and text encoders with CLIP weights, improving GZSD HM and ZSD AP_{50} by 3.1 and 3.6 points, respectively. Based on this initialization, Table VI directly compares the two alignment strategies. Learning both visual- and text-side projections yields a GZSD HM of 5.6% and a ZSD AP_{50} of 7.0%. When the additional text-side projection is removed, and only the visual-side projection is learned, the two metrics improve by 3.2 and 2.8 points, respectively. These results show that the proposed image-to-text alignment improves novel-class detection under the aerial zero-shot setting.

2) Ablation on Image–Text Collaboration Strategy: As illustrated in Table VI, introducing TG-FE or TG-QE individually yields notable performance improvements across both experimental settings. When both modules are employed together across the encoder and decoder, the GZSD HM and ZSD AP_{50} reach 13.1% and 14.3%, respectively. Furthermore, when the VG-TR is introduced on top of TG-FE and TG-QE, the model obtains the highest GZSD HM of 13.9%, validating that enriching text embeddings with context-aware visual cues further boosts image–text alignment and improves generalization to novel classes. These results highlight the complementary benefits of the TG-FE, TG-QE, and VG-TR modules and confirm the overall effectiveness of the proposed image–text collaboration design.

TABLE VII

ABLATION STUDY ON BACKGROUND-INTERFERENCE SUPPRESSION

w/ suppress		w/o suppress		GZSD	ZSD
TG-FE	TG-QE	TG-FE	TG-QE	HM	AP_{50}
				8.8	9.8
		✓		9.3	11.7
✓				10.4	12.3
✓			✓	11.6	12.5
✓	✓			13.1	14.3

3) Ablation Study on Interference Suppression: Table VII demonstrates the effectiveness of the interference suppression branch. Experiments show that using text information as guidance to enhance features in either the encoder or decoder is effective. Compared to the baseline, this approach yields improvements of 0.5% and 1.2% in HM of AP_{50}, respectively. Introducing the suppression branch further improves detection accuracy. For instance, for the TG-QE module, introducing the suppression branch further improves the HM of AP_{50} by 1.5%.

4) Visual Analysis on the Impact of the Proposed TG-FE: Fig. 5 visualizes the impact of the TG-FE module on the features. Specifically, we visualized the correlation between visual features and the text prompt before and after the introduction of the TG-FE module. The first row corresponds to cases with larger objects, while the second row corresponds to cases with smaller objects. Comparing the results in the second and third columns, it is evident that after applying the TG-FE module, the correlation between visual features and the text prompt is significantly enhanced. In particular, in the second-row example where the ship is relatively small, the initial visual features exhibited a vague correlation spread across various parts of the image. After applying TG-FE, the ship's features became much more prominent, and the background was effectively suppressed. As shown in the third row, if the text prompt does not match the image content, the entire image is suppressed. This visualization confirms the effectiveness of TG-FE in enhancing class-relevant features and suppressing background. The roles of VG-TR and TG-QE are analogous: TG-FE and VG-TR refine the encoder representations, while TG-QE refines the decoder queries, jointly strengthening image–text alignment in ITC-OVAD.

V. DISCUSSION

A. Per-Class Performance Analysis

To provide a finer-grained understanding of ITC-OVAD's behavior on novel classes, we report the class-wise AP_{50} of novel categories on DIOR, DOTA, and xView under both ZSD and GZSD settings in Table VIII. From the results, ITC-OVAD tends to achieve relatively larger improvements on categories with compact and visually distinctive characteristics. For example, under ZSD, basketball court on DIOR improves from 18.2 to 40.8, helicopter on DOTA from 5.0 to 16.6, and bus on xView from 5.7 to 16.7. In contrast, spatially

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TABLE VIII

CLASS-WISE AP_{50} COMPARISON ON NOVEL CATEGORIES ACROSS THREE AERIAL DATASETS UNDER GZSD AND ZSD SETTINGS. RESULTS ARE REPORTED USING THE REDEFINED BASE/NOVEL SPLIT (*) TO AVOID NOVEL-CLASS LEAKAGE

		DIOR				DOTA				xView												
Setting	Method	airport	bask. f.	gr. tra. f.	windmill	bask. c.	heli.	gr. tra. f.	swim. pool	heli.	bus	pic. truck	tru. tra. w/ box tra.	mar. vessel	motorb.	barge	reach stacker	mobile crane	scraper	excavator	ship. cont.	
GZSD	LAE-DINO	1.7	9.3	9.4	5.7	12.0	0.2	6.8	0.2	0.1	7.3	0.1	7.7	14.4	16.1	3.0	0.1	0.5	0.4	6.1	0.0	
	ITC-OVAD (Ours)	0.3	23.1	3.7	3.3	6.6	2.7	2.1	0.3	7.1	6.3	0.4	12.3	18.5	14.3	18.5	0.3	0.4	0.7	5.2	0.0	
ZSD	LAE-DINO	4.7	18.2	10.4	8.3	10.7	5.0	15.7	0.6	0.7	5.7	0.5	47.6	14.6	20.4	3.2	0.5	0.9	0.5	12.1	0.1	
	ITC-OVAD (Ours)	0.6	40.8	9.3	5.7	13.7	16.6	7.0	1.1	18.0	16.7	1.1	26.3	21.7	24.6	25.0	2.2	0.8	1.9	9.0	0.3	

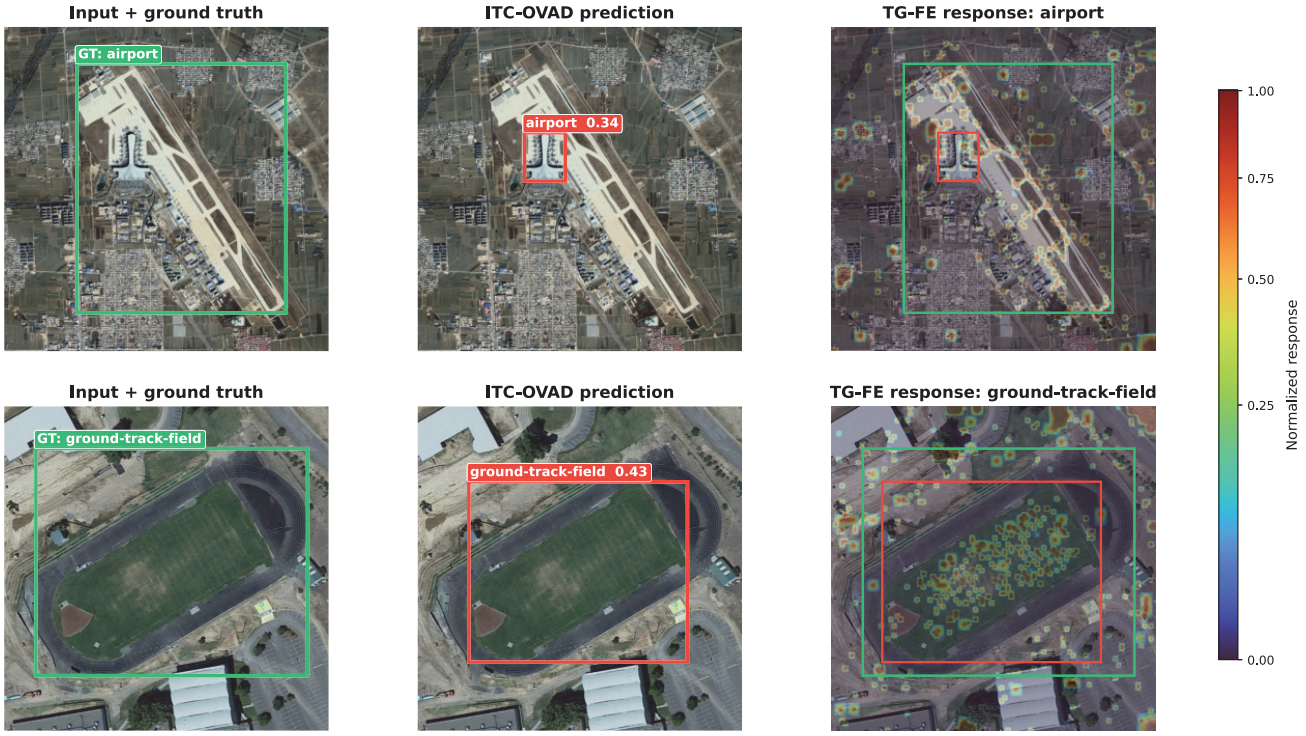


Fig. 6. Target-class TG-FE responses for spatially diffuse scene-level categories under the DIOR Protocol 2 ZSD setting.

diffuse scene-level categories such as airport and ground-track-field remain relatively difficult.

To further analyze this behavior, Fig. 6 visualizes the target-class TG-FE responses for airport and ground-track-field under the DIOR Protocol 2 ZSD setting. In the airport example, the response is distributed across the runway, apron, buildings, and other local structures. Although the predicted category is correct, the bounding box covers only a small part of the annotated region. For ground-track-field, the response is mainly concentrated on the track and inner field rather than the complete scene-level annotation. These examples show that local cues can support category recognition, while the model may still have difficulty determining the full spatial extent of scene-level categories.

B. Analysis of False Positives

To examine the high-recall/low-AP pattern, we analyze all predictions with a score of at least 0.3 on the complete DIOR, DOTA, and xView evaluation sets. Each false positive is assigned to one of four sources. Semantic confusion denotes an IoU of at least 0.5 with a differently labeled GT object. Localization error denotes a maximum IoU between 0.1 and 0.5. A duplicate is a repeated same-class detection of an already matched object. Pure background requires a maximum IoU of at most 0.1 with every GT box.

As shown in Table IX, strict ZSD and GZSD exhibit different error patterns. Under ZSD, pure-background predictions account for 72.8%, 84.2%, and 76.3% of all false positives on DIOR, DOTA, and xView, respectively. Under GZSD, 78.3%–96.4% of false positives overlap differently labeled objects, while pure background accounts for only 1.6%–16.0%. Thus, background responses are the main source of false positives under strict ZSD, whereas GZSD is primarily affected by confusion between visually similar categories.

Fig. 7 gives a class-level view using DIOR. Fig. 7(a) shows the ZSD precision–recall curves. Fig. 7(b) and Fig. 7(c) shows column-normalized false-positive sources under ZSD

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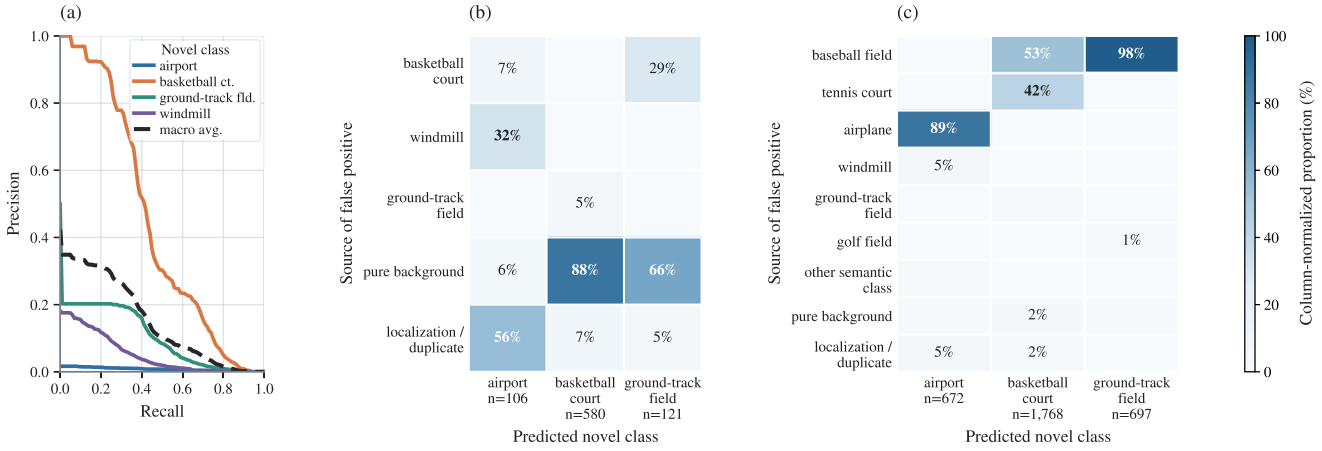


Fig. 7. Class-level diagnostics from a representative DIOR run. (a) ZSD precision–recall curves. (b) and (c) Column-normalized false-positive source matrices under ZSD and GZSD, respectively. Each column summarizes the false positives predicted as one novel class; semantic rows denote the overlapping ground-truth class, and n denotes the number of false positives.

TABLE IX

PERCENTAGE DISTRIBUTION OF FALSE-POSITIVE SOURCES. SEM. CONF., BACKGROUND, LOC., AND DUP. DENOTE SEMANTIC CONFUSION, PURE-BACKGROUND PREDICTION, LOCALIZATION ERROR, AND DUPLICATE DETECTION, RESPECTIVELY

Setting	Dataset	Sem. conf.	Background	Loc.	Dup.
ZSD	DIOR	12.5%	72.8%	14.7%	0.0%
	DOTA	7.5%	84.2%	8.0%	0.3%
	xView	20.5%	76.3%	3.2%	$< 0.1\%$
GZSD	DIOR	96.4%	1.6%	2.0%	$< 0.1\%$
	DOTA	91.1%	6.4%	2.5%	0.0%
	xView	78.3%	16.0%	5.6%	$< 0.1\%$

and GZSD, respectively, where each column contains all false positives predicted as one novel class. Under ZSD, pure background accounts for 88% and 66% of the basketball-court and ground-track-field false positives, respectively. Under GZSD, 89% of airport false positives overlap airplane GT, basketball-court false positives mainly overlap baseball-field (53%) and tennis-court (42%) GT, and 98% of ground-track-field false positives overlap baseball-field GT. These findings also point to two directions for future work: stronger background suppression under ZSD and better separation of visually similar categories under GZSD.

TABLE X

PER-MODULE INFERENCE OVERHEAD ON AN NVIDIA RTX 4090 GPU WITH AN 800 × 800 INPUT AND BATCH SIZE 1. † INDICATES THAT THE SHARED TG-QE MODULE IS CALLED AT ALL SIX DECODER LAYERS; ITS FLOPS AND LATENCY INCLUDE ALL SIX CALLS

Module	Params (M)	FLOPs (G)	Memory (MB)	Latency (ms)
TG-FE	3.51	38.59	14.04	1.76
VG-TR	4.26	57.84	17.04	2.19
TG-QE†	3.51	8.86	14.04	4.17 (0.69×6)

TABLE XI

CONTROLLED TRAINING-STAGE COMPARISON WITH AND WITHOUT TG-QE. GRADIENT NORMS ARE MEASURED BEFORE CLIPPING AND NORMALIZED TO THE SETTING WITHOUT TG-QE

Metric	Without TG-QE	With TG-QE	Change
Time/iter. (ms)	527.75	547.38	+3.72%
Peak memory (MiB)	9,348	9,670	+3.44%
Loss mean±std.	18.61 ± 3.27	18.47 ± 3.22	−0.75%
Relative grad. norm (mean/P95)	1.000/1.000	0.756/0.879	−24.4%/ − 12.1%
NaN/Inf	0	0	–

TABLE XII

COMPARISON USING THE SAME BACKBONE AND INPUT RESOLUTION

Method	FPS	DIOR AP_{50}	DOTAv2 mAP	LAE-80C mAP
Grounding DINO-CLIP	14	84.5	46.4	15.4
LAE-DINO-CLIP	13	81.9	44.6	19.1
ITC-OVAD	**34**	**87.7**	**53.8**	**23.7**

C. Module Computational Overhead and Training Stability

We evaluate the computational overhead of the proposed modules at inference and further examine whether repeatedly applying TG-QE affects training stability.

1) Inference-Stage Overhead: Table X reports the overhead of each module during inference. TG-QE uses a single 3.51M-parameter module shared across the six decoder layers. Its reported 8.86 G FLOPs and 4.17-ms latency therefore include all six calls. Together, the three modules add 11.28M parameters and approximately 8.12-ms latency.

2) Training-Stage Overhead and Stability: Because TG-QE is invoked at all six decoder layers, we further examine whether its repeated use increases training overhead or affects gradient stability. We compare two otherwise identical variants with and without TG-QE. Both variants start from the same checkpoint and process identical augmented mini-batches. We use FP32, 800 × 800 inputs, and a batch size of 2 per GPU. Statistics are collected over 400 optimization iterations, and all gradient norms are measured before clipping.

As shown in Table XI, enabling TG-QE increases the iteration time by 3.72% and peak memory by 3.44%, while the mean training loss remains comparable (18.47 versus 18.61). The mean and P95 preclipping global gradient norms are 24.4% and 12.1% lower, respectively, and neither variant

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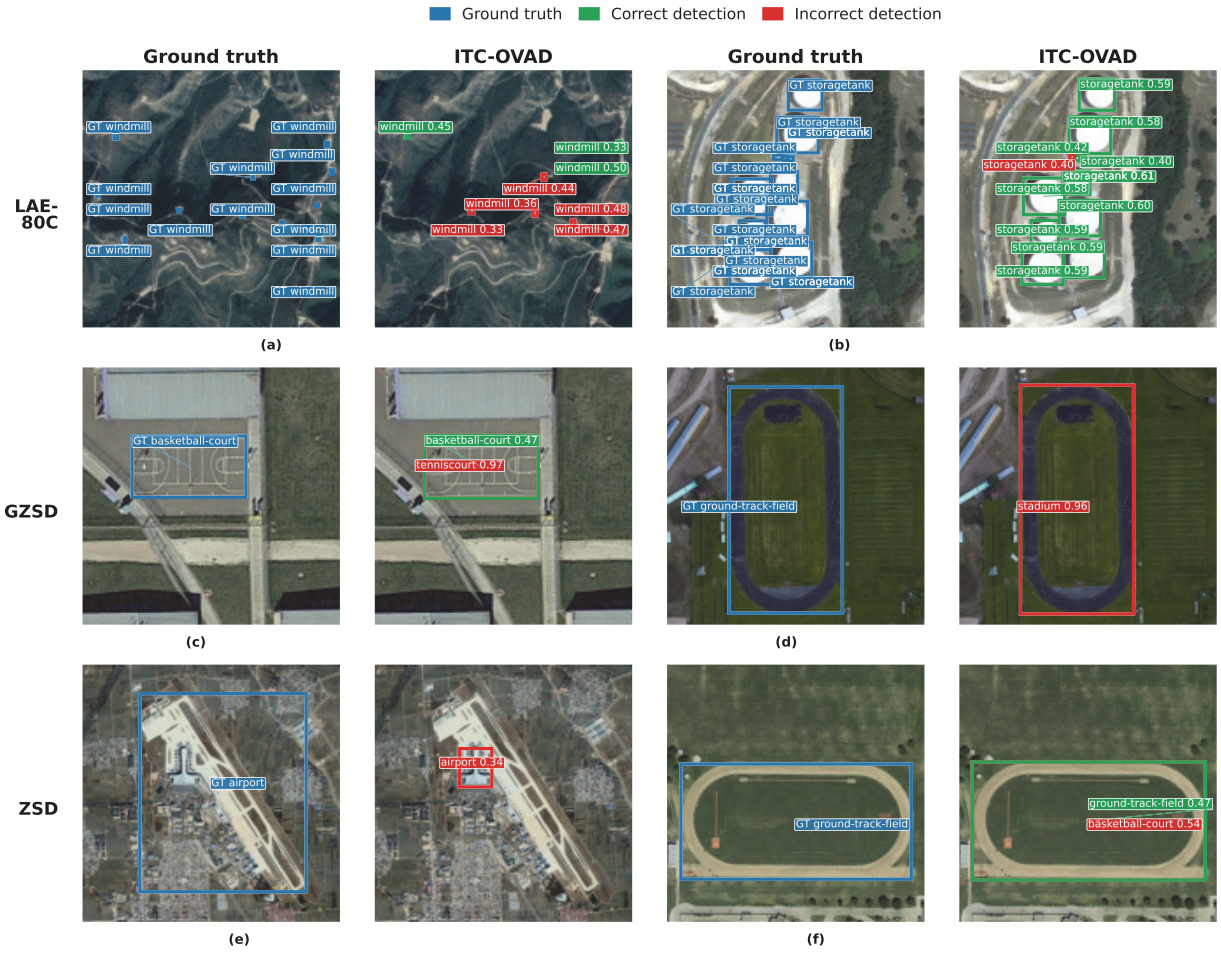


Fig. 8. Representative failure cases on LAE-80C and under the GZSD and ZSD protocols. The three rows show dense small-object errors, novel-to-base confusion, and ZSD localization or novel-class confusion, respectively. (a) Dense small windmills. (b) Dense small storage tanks. (c) Novel basketball court → base tennis court. (d) Novel ground-track-field → base stadium. (e) Airport part–whole localization. (f) Novel-class duplicate hypothesis.

TABLE XIII
DENSITY-BASED PERFORMANCE ANALYSIS UNDER THE OPEN-VOCABULARY SETTING. IMAGES ARE PARTITIONED BY THE NUMBER OF GT OBJECTS PER IMAGE. ITC-OVAD USES 500 OBJECT QUERIES

Dataset	Density Bin	Imgs	mAP	AP_{50}
LAE -80C	[0, 50)	3170	23.4	34.5
	[50, 200)	351	17.8	27.1
	[200, 500)	61	11.4	21.0
	[500, +∞)	10	0.7	1.4
DOTA	[0, 50)	735	56.2	83.0
	[50, 200)	108	61.4	84.2
	[200, 500)	26	39.3	56.8
	[500, +∞)	5	29.6	48.5
DIOR	[0, 50)	579	69.9	87.7
	[50, 200)	7	67.0	88.9

produces NaN or Inf values. These results show that the six TG-QE calls introduce modest training overhead without causing gradient amplification or numerical instability.

D. Controlled Comparison Under Matched Backbones

To further isolate the influence of the backbone configuration and input resolution, we compare ITC-OVAD with adapted versions of Grounding DINO and LAE-DINO under the same setting. All three methods use a CLIP-RN50 visual backbone, a frozen CLIP text encoder, and an input resolution of 800 × 800, and are fine-tuned on LAE-1M. The original feature interaction design, decoder, and detection head of each method are otherwise retained, and all methods use the same evaluation protocol.

As shown in Table XII, ITC-OVAD outperforms Grounding DINO-CLIP by 3.2, 7.4, and 8.3 points on the three datasets, respectively, and LAE-DINO-CLIP by 5.8, 9.2, and 4.6 points. ITC-OVAD also achieves a substantially higher inference speed. These results confirm that its performance and efficiency advantages remain when the visual backbone, text encoder, and input resolution are matched.

E. Limitations and Future Directions

1) Limitations: While ITC-OVAD achieves strong performance across three benchmark tasks, several limitations remain.

1) High recall but low AP on novel classes. As shown in Tables II and III, all compared detectors exhibit high recall but low AP on DIOR, DOTA, and xView. Our false-positive analysis further shows that the dominant

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source of error differs between the two evaluation settings. In strict ZSD, most false positives arise from background regions, whereas under GZSD, they mainly result from confusion between visually similar categories.

2) Performance degradation in dense scenes. As shown in Table XIII, detection performance degrades as scene density increases, since dense scenes contain a high proportion of small, closely packed objects that exacerbate both localization difficulty and interobject confusion. Accurate detection in extremely dense aerial scenes remains a significant challenge.
3) Lack of fine-grained textual annotations. Current open-vocabulary aerial benchmarks provide only category-level labels, and the lack of finer-grained semantic descriptions (e.g., phrase- or sentence-level) limits the expressiveness of textual guidance and hinders the construction of more robust image–text alignment.
4) Zero-shot performance fluctuations. Under zero-shot settings, mild performance fluctuations are observed across runs, which we attribute to the sparsity of textual prompts for novel classes, the long-tailed category distribution, and inherent stochasticity in training. We report the average results over three independent runs to improve evaluation stability.

Fig. 8 further illustrates the above limitations through representative failure cases. In dense LAE-80C scenes, small and closely spaced objects are sometimes missed or localized inaccurately. The GZSD examples show confusion between novel classes and visually similar base classes, while the ZSD examples show incomplete localization of scene-level targets and competing predictions between visually related novel classes. These cases provide visual examples of the localization and category-confusion errors identified in the preceding analyses.

2) Future Directions: Based on the above analysis, we discuss future directions from two aspects.

1) *Performance Improvement:* Constructing larger-scale detection or grounding datasets with fine-grained textual annotations could strengthen image–text alignment and improve zero-shot generalization.
2) *Deployment Acceleration:* ITC-OVAD is built upon RT-DETR, which natively supports ONNX export and TensorRT deployment. In addition, PaddleDetection provides automated quantization compression for RT-DETR via the PaddleSlim Auto Compression Toolkit (ACT) [55], offering a practical deployment path. Furthermore, given the high redundancy of background tokens in aerial imagery, text-guided token pruning that leverages the foreground probability to discard low-score visual tokens before the decoder represents another promising direction for inference acceleration.

VI. Conclusion

In this article, we present ITC-OVAD, a lightweight detector for open-vocabulary aerial object detection built around two main components. The image-to-text alignment component incorporates class semantics into the detector through an image-to-text alignment loss that connects visual objects with class embeddings, eliminating the predefined category constraints of traditional detectors. The image–text collaboration component employs bidirectional modulation between visual and textual modalities through three lightweight modules: TG-FE and VG-TR in the encoder, and TG-QE in the decoder. These modules enable ITC-OVAD to focus on class-relevant features, suppress background interference, and effectively exploit textual guidance. Moreover, by avoiding computationally intensive modules, ITC-OVAD inherits the efficiency of its base detector and meets the real-time demand of aerial detection. In future work, we will extend textual guidance from word-level cues to phrase- and sentence-level descriptions, providing richer semantic information to enhance open-vocabulary detection capabilities and broaden the model's applicability.

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